Supplementary Material

This document contains the supplementary materials for the article Adversarial Random Forests for Omics Synthesis.

1 Benchmark studies

1.1 Downstream clustering

Figure S1 shows the impact the empirical ARI and MMD for different values the of chunk sizes in the h-ARF algorithm. Because the effect of the chunk size was similar for all performance metric, we show only the empirical MMD for distributional similarity metric and the empirical ARI and downstream analysis utility. In figure S2, we show the runtime of h-ARF and ARF for chunk variation.

1.2 Downstream prediction

As the tendance was the same for MMC, we only show the empirical AUC values against the chunk size variation (figure S4). Analaguously, we show the impact of chunk size on MMD only, as the behaviour was similar with the other distribution metrics (S3).

Figure S1: Empirical A) ARI and B) MMD are shown for downstream clustering datasets for different h-ARF chunk sizes and for ARF.

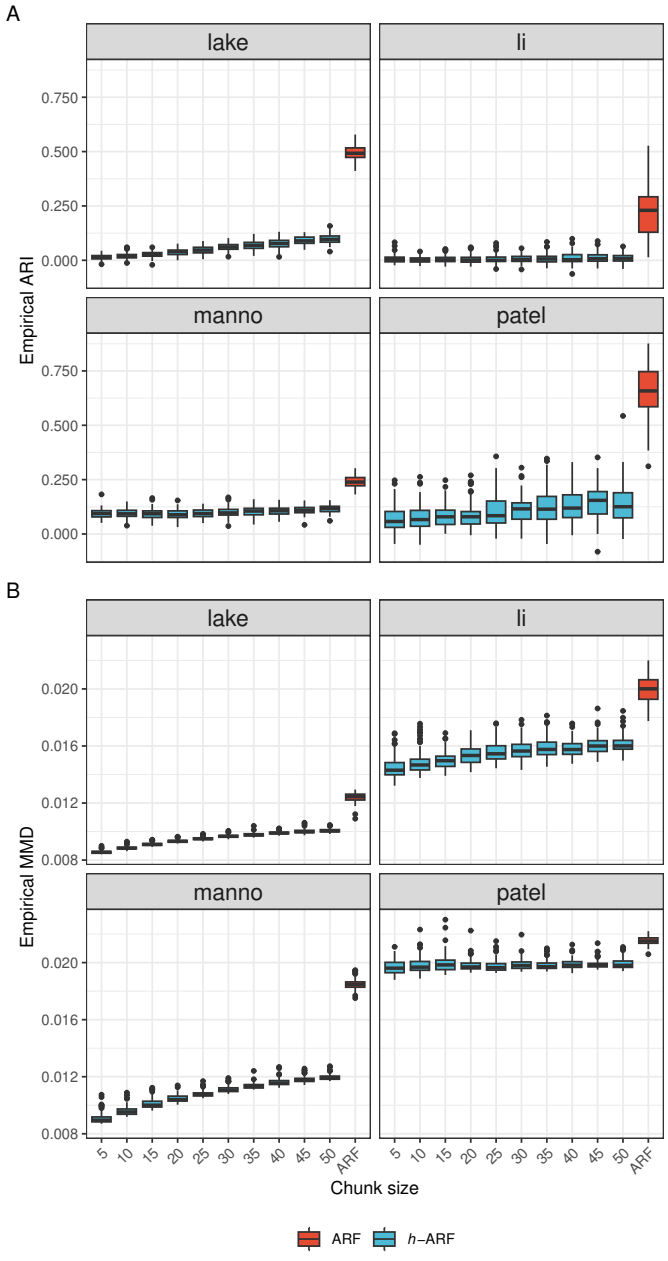


Figure S2: The log10-transformation (for better visualization) of empirical runtimes are shown for downstream clustering datasets for different h-ARF chunk sizes and for ARF.

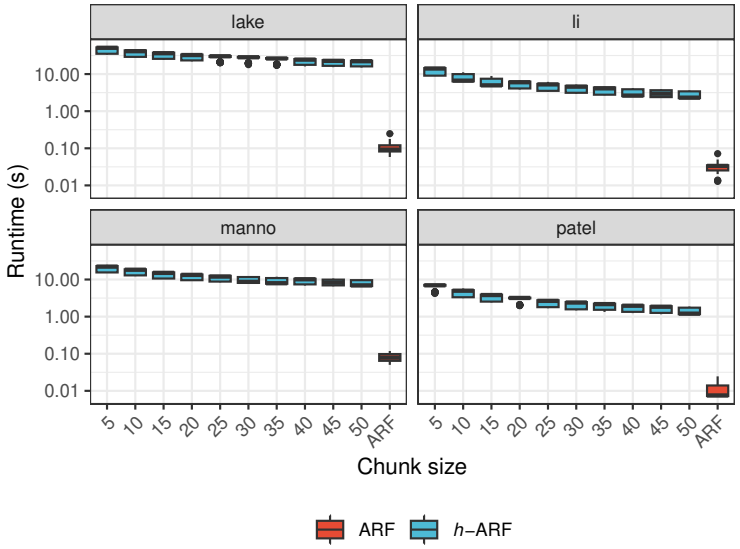


Figure S3: Empirical MMD values are downstream prediction datasets for different h-ARF chunk sizes and for ARF.

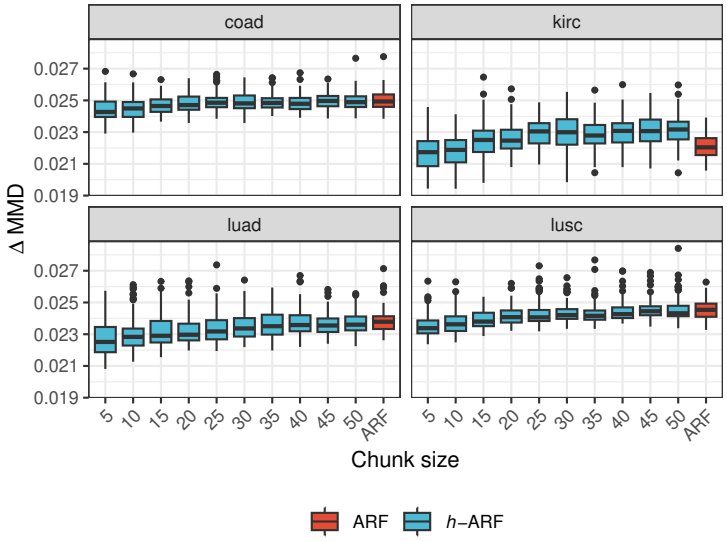


Figure S4: Empirical AUC values are shown for A) RF and B) Lasso classifiers for downstream prediction datasets for different h-ARF chunk sizes and for ARF.

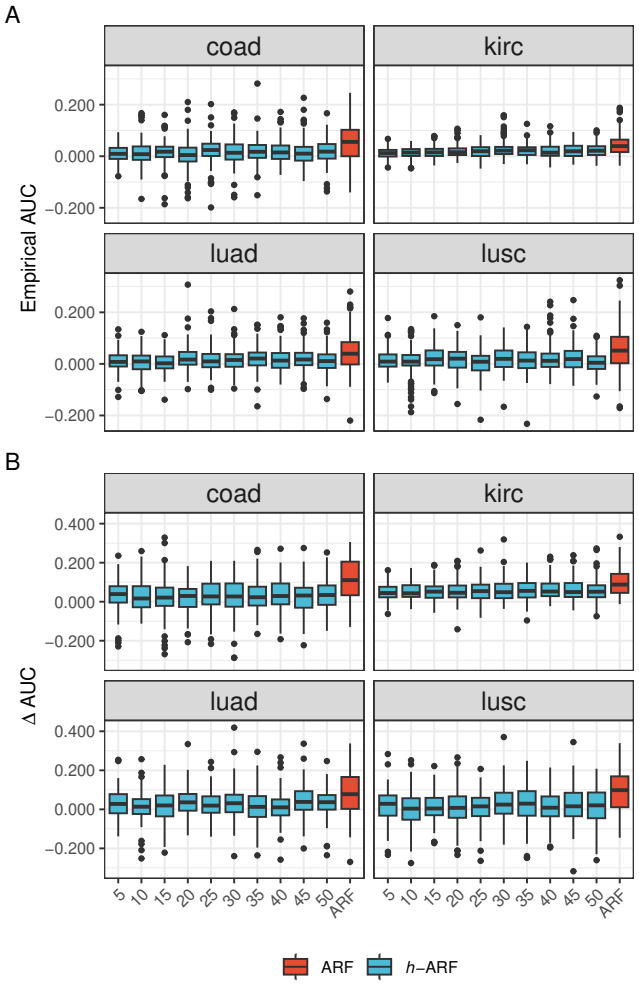


Figure S5: Empirical runtimes for downstream prediction datasets for different h-ARF chunk sizes and for ARF.

